

Shubham Kumar

B.Tech Computer Science Engineering (3rd Year)

7870841473 | shubhamkm07@gmail.com | linkedin.com/in/shubham-kumar | github.com/myselfshubham1

Education

ITM University

B.Tech – Computer Science Engineering

CGPA: 8.3/10.0

2023 – 2027

Technical Skills

Languages: C, C++, Java, Python, SQL

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems

Backend / Systems: C++ STL, File Handling, MySQL, REST APIs, Postman, JWT Authentication

Frameworks / Libraries: React.js, Next.js, Express.js, Node.js, TypeScript

Database: MySQL, MongoDB, Mongoose, PostgreSQL, Prisma ORM

Deployment / Tools: Vercel, Neon Serverless PostgreSQL, Git, Vite

AI / ML: NumPy, Pandas, Scikit-learn, YOLOv8, OpenCV

Web: HTML, CSS, JavaScript, React

Relevant Coursework

Data Structures & Algorithms · Object-Oriented Programming · Database Management Systems · Operating Systems · Computer Networks · Software Engineering · Discrete Mathematics · Machine Learning

Experience

Software Development Intern – Blog Post Management System

Industrial Training

Intellige IT Studio

Java, MySQL, REST APIs

- Completed industry-oriented software development training and built a fully functional blog post management system using **Java**, **MySQL**, **HTML**, **CSS**, **REST APIs**, and **Postman**.
- Designed and implemented **REST**-based backend APIs for blog and user management supporting full CRUD operations (create, update, search, delete).
- Implemented user and admin roles with role-based access control for secure content management.
- Enabled tracking of creation and update timestamps for blogs, ensuring data audit capabilities.

Projects

MCP Server – AI-Driven 3D Model Generation | Python, Socket Programming, Blender, Cursor AI

- Designed and implemented a custom **MCP server** using socket-based communication for real-time tool integration.
- Integrated **Blender** and **Cursor AI** to enable 3D object generation using natural language prompts.
- Built a scalable, real-time system capable of connecting multiple tools simultaneously.
- Enabled full customization and modification of 3D models based on user input.
- Implemented robust error handling and reconnection logic to maintain stable socket connections during long-running generation tasks.
- Structured the MCP protocol interface to be extensible, enabling future AI tools to plug into the server with minimal integration overhead.

CivicChain – Civic Intelligence Platform | React, TypeScript, Vite, Node.js, Express, PostgreSQL, Prisma

- Built a full-stack civic-tech web application enabling citizens to report geo-tagged civic issues, track resolution status, and monitor ward-level official accountability and public spending.
- Architected the frontend with **React**, **TypeScript**, and **Vite**, integrating **Mapbox GL JS** for interactive WebGL mapping with heatmaps, 3D buildings, and custom geographic clustering.
- Designed a **PostgreSQL** database with **Prisma ORM** to model issues, ward officials, and users with relational mappings, replacing the initial **MongoDB** prototype for stronger relational integrity.
- Implemented secure **JWT**-based authentication with role-based access and a dual-mode auth system supporting both local and API-based sign-in.
- Developed a **REST API** using **Express** and **Node.js** with endpoints for issue reporting, community upvoting, and a ward-official accountability dashboard (NetaMeter) with trust and resolution scores.
- Deployed the application on **Vercel** with a live production build, using **Neon serverless PostgreSQL** for the database layer.

Mini Projects

To-Do Application – Full-stack task management app with CRUD functionality

Chatbot – Interactive conversational bot using rule-based/NLP techniques

Calculator – Multi-function calculator application with scientific operations

Profile Calculator – Profile analytics and scoring tool

Number Guessing Game – Interactive CLI game with randomized logic and difficulty levels

Achievements & Hackathons

- Winner – Hackathon conducted by ITM University
- Finalist – “FinShield” Hackathon conducted by Bank of India; presented solution at IIT Hyderabad
- Participant – Smart India Hackathon (SIH) and actively participated in multiple inter-college hackathons